CPU overhead and frame freshness in a robot perception pipeline with five cameras

Structured abstract

Purpose To quantify how camera transport and conversion configurations trade processor demand against receiver frame freshness in a deployed robot perception pipeline.

Design/methodology/approach A retrospective reanalysis used 30 complete, fixed-order runs from five historical batches on one robot. Each batch contained three shared-memory and three UDP runs with five camera roles. CPU measurements were harmonized to 60–240 seconds. Frame-age aggregates were complemented by exact preview-frame joins, subscriber diagnostics and audits of primary and additional long-duration archives.

Findings Shared-memory runs used 9.47–20.02 percentage points less camera-process CPU than their adjacent UDP runs, with 100% representing one core. An exploratory interpolation check retained this direction under locally linear drift. The hardware-conversion batch averaged 29.41% and 39.77% CPU, respectively, while its worst role/run raw-frame p95 ages were 4.66 and 3.96 ms. Matched preview p95 delays remained near 70 ms despite millisecond-scale raw-image age. Exploratory allocator diagnostics exposed substantial subscriber overhead. The primary long-duration record lost node/RSS observations. A separate 24-hour archive passed its observation checks but retained an RSS-growth warning; preview-conversion checks also failed.

Originality The study links process accounting, matched preview delays and observer overhead in an auditable deployment case, while separating monitoring completeness from memory behavior and retaining adverse conversion evidence. It supplies reproducible source tables and bounded selection examples without equating pipeline metrics with policy success.

Keywords robot perception; shared memory; UDP; frame freshness; CPU overhead; resource allocation

Article classification Original Article; Research paper

1 Introduction

A robot camera pipeline must serve several consumers with different timing and resource requirements. Raw images support perception, encoded streams support recording, and reduced-rate previews support human observation. These consumers can share capture, conversion and transport resources. Consequently, a configuration that reduces one process's CPU demand can alter frame age, queue occupancy or another consumer's behavior. The middleware architecture of Robot Operating System 2 (ROS 2) makes such configuration choices central to deployed robotic systems (Macenski *et al.*, 2022).

Processor capacity is shared with state estimation, planning and learned policies. Camera savings create potential computational headroom, but do not establish faster decisions, higher task success or more accurate observations. Similarly, approximately 30 frames per second does not establish sufficiently recent frames.

Communication and tracing methods still require deployment-specific validation of the complete consumer configuration. This study examines an existing five-camera robot pipeline in which shared-memory and User Datagram Protocol (UDP) subscribers were compared across successive

engineering batches. The retained records also contain hardware-conversion checks and stability tests, enabling favorable short-run measurements to be assessed alongside adverse evidence.

The research question is whether lower camera-process CPU demand coincided with equal receiver freshness and adequate evidence of sustained operation in this deployment. First, the analysis reconstructs the resource and frame-age trade-off across 30 complete executions, while preserving the run as the unit of technical repetition. Second, exact preview-frame joins and historical subscriber diagnostics expose timing boundaries and observer overhead that aggregate throughput obscures. Third, it supplies anonymized source tables and a reproducible screening procedure for explicit CPU and frame-age budgets.

Historical collection was neither randomized nor factorial. This empirical systems case characterizes the retained deployment; it does not isolate a causal effect of shared memory or hardware conversion.

2 Related work

2.1 Communication and memory management

ROS 2 provides a configurable communication architecture spanning processes and machines (Macenski *et al.*, 2022). Kronauer *et al.* (2021) showed that middleware choice and software-stack overhead affect end-to-end latency. Their work supports measuring an actual application configuration rather than treating middleware labels as performance guarantees.

Zero-copy communication further depends on ownership, message representation and memory lifetime. Agnocast addresses unsized message types in ROS 2 interprocess communication (Ishikawa-Aso and Kato, 2025). Subsequent work examines reclamation and metadata-management trade-offs across publisher/subscriber systems (Ishikawa-Aso *et al.*, 2026). Hardware heterogeneity introduces additional device-memory transfers. Hazcat addresses these operations across CPU, GPU and FPGA components (Bell *et al.*, 2023). These contributions motivate examining the conversion and consumer paths surrounding communication. The present dataset cannot establish that its shared-memory configuration removes every image copy.

2.2 Freshness and application performance

Information freshness differs from both throughput and packet latency. Age-of-information research evaluates the recency of source updates available to a receiver over time (Kaul *et al.*, 2012; Yates *et al.*, 2021). Here, the available camera metric is timestamp-to-receipt age evaluated at message arrivals. It is related to freshness, but it is not the continuous-time average age of information.

Perception optimization also requires quality measurements. Park *et al.* (2025) evaluated accuracy and latency for open-vocabulary perception models on an embedded platform, including unfavorable precision configurations. Their application-level evaluation differs from the transport and conversion measurements available here. A low-cost image-delivery path remains insufficient evidence for detection accuracy or robotic task success.

2.3 Observation and retrospective measurement

The ros2_tracing framework provides low-overhead runtime instrumentation for distributed robotics (Bédard *et al.*, 2022). Message-flow analysis further connects causal links across processing stages (Bédard *et al.*, 2023). These studies explain the value of preserving timing boundaries and causal identifiers. Our files support CPU reconstruction and exact preview-frame joins, while ROS per-message age samples are absent. Accordingly, the analysis separates quantities that can be recomputed from those available only as recorded aggregates.

3 Materials and methods

3.1 Deployment and evidence provenance

The records were generated on 13–14 September 2026 (UTC) by a single robot's embedded camera system. Five roles were active: head, chest, left wrist, right wrist and an auxiliary wrist camera. Interface snapshots show 848 × 480 RGB8 raw images. Diagnostic snapshots identify YUYV capture, H.265 recording-stream encoding and H.264 preview encoding. The raw stream operated near 30 Hz; decoded preview streams operated near 15 Hz. These are concurrent consumer branches with different rates.

Historical CPU logs report 14 online processors; the later device roster identifies an NVIDIA Jetson AGX Thor Developer Kit. Historical power mode, clocks, thermal state and operating-system image remain unknown. Collection scripts select ROS 2 Jazzy and Cyclone DDS. Associated installation records report versions 0.45.91 for F and 0.46.90 for V, but exact per-run executable hashes and commits are absent. Batches therefore remain separate observational strata.

The primary archive contains 1,059 copied files (271,022,182 bytes), including two frame traces. SHA-256 hashes, sizes and modification times document transfer consistency, without proving an atomic historical snapshot. Subsequent fleet-wide read-only retrieval identified additional long-duration and capture-loop archives. Their differing workloads remain contextual cohorts (Supplement S9–S10). No collection program or robot control was executed for this reanalysis.

[Table I near here]

3.2 Cohort reconstruction and collection design

Forty-two summaries were located. Thirty short runs belonged to five complete batches, each containing three shared-memory (SHM) and three UDP executions. The original directory labels were baseline, baseline2, final, baseline3 and nvmm, represented here as B1, B2, F, B3 and V. Labels identify retained batches, not verified equivalence of software versions. Nine additional short-run summaries and three stability summaries were retained as contextual evidence.

The collection script alternated SHM then UDP for repetitions 1–3, with a short pause between executions. Within a batch, the camera publisher remained active while subscribers were recreated. Each run requested 60 seconds of warmup and 180 seconds of measurement. One camera probe subscribed to the raw and encoded ROS topics for all roles. Five WebRTC HTTP Egress Protocol (WHEP) probes decoded previews concurrently. CPU sampling occurred approximately once per second, with diagnostic snapshots approximately every ten seconds.

The SHM probe ran with elevated privileges and a shared-memory Cyclone DDS profile; UDP ran as the ordinary user with that profile unset. Thus, the comparison evaluates the deployed configuration bundles. It does not isolate transport from privilege, memory-registration or configuration differences. The workload is also an instrumented camera-consumer workload. The presence of an image branch called inference does not establish that a learned policy was running.

3.3 Resource and freshness outcomes

Process CPU usage is expressed relative to one processor core, so 100% represents one fully occupied core. System CPU uses the same scale summed over online processors. The primary statistic is each run's mean camera-process CPU over elapsed time [60, 240) seconds after the first sampler timestamp. This common interval excludes subsequent cleanup samples retained by some

historical summaries. A total of 5,399 samples fell in the primary windows; individual runs contained 179–181 samples because sampling was not perfectly periodic.

For camera role c and message j, receiver frame age is $A(c,j) = r(c,j) - s(c,j)$, where r denotes receipt time and s denotes the message timestamp. Recorded per-role p50, p95 and p99 values describe the full approximately 240-second probe interval, including startup. Raw per-message ages were not retained, so warmup cannot be removed retrospectively. The worst role/run p95 is the maximum of the 15 recorded p95 values within a batch/transport cell. It is a conservative observed summary, not a pooled p95 or a bound on all future frames. Timestamp mapping also prevents interpreting this metric as an independently calibrated optical-exposure-to-action latency.

Resident set size (RSS) is converted from KiB to MiB. Short-run RSS trajectories are plotted through teardown instead of interpreting a final sample in isolation. Diagnostic drop counts are cumulative. For each role, increments are calculated between the first and last diagnostic observations inside the CPU window, with intermediate decreases flagged as resets. Counter intervals are shorter than the complete CPU window and vary with snapshot timing. These counters describe branch behavior; they are not measured task failures or confirmed missing dataset frames.

Preview frames are joined to retained publisher traces by session identifier and 32-bit RTP timestamp. Same-host monotonic timestamps define capture-marker-to-output, capture-marker-to-receipt and capture-marker-to-decoded delays. Duplicate keys, role mismatches and temporal-order violations are checked. Full records and a 60–230-second window after each first decoded frame are analyzed separately. These software markers do not measure optical exposure or display latency. Unmatched frames remain explicit (Supplement S7).

Eight additional Cyclone DDS subscriber diagnostic conditions test consumer selection and jointly raised allocator thresholds. Each condition has one historical execution, with debugger attachment and fixed order; five earlier Fast DDS conditions are retained separately. CPU profiles span 60 or 45 seconds after 30-second warmup. These are exploratory diagnostics, not additional primary repetitions (Supplement S8).

3.4 Analysis and robustness checks

Adjacent SHM/UDP executions with the same repetition index form descriptive technical pairs. Primary effects are absolute CPU differences, relative differences and observed ranges. Cameras and one-second samples are dependent observations within a single system. No population-level significance test, independent-robot sample size or randomized causal estimate is claimed. Observed ranges describe technical repeatability, and leave-one-pair-out contrasts assess sensitivity to any one pair. Neither provides population uncertainty. An exploratory order check interpolates neighboring SHM run means at each intervening UDP measurement centroid within a batch. It evaluates locally linear common drift while leaving nonlinear drift, carryover and privilege confounding unresolved (Supplement S11).

Robustness checks use warmup cutoffs of 30, 90 and 120 seconds, the original unbounded post-warmup summaries, and time-weighted CPU averages. Archived short-run CPU means are independently recomputed using the retained summarizer. Stability records use ordinary least-squares (OLS) and Theil–Sen RSS slopes, explicit missingness and alternative warmup cutoffs. Two additional nominal 24-hour archives are audited separately using probe coverage, host-monitor gaps and periodic recording checks. RSS observations are never interpolated. Retained source hashes and measurement definitions are checked before interpreting these additional records.

Finally, a post hoc screening example minimizes measured CPU subject to an illustrative raw-frame-age budget. Its age statistic is the worst observed role/run p95, and the CPU constraint uses the cell's mean run CPU. These analyst-selected budgets are not task deadlines or safety requirements. Screening is separated from sustained-operation qualification, which also requires complete stability observations and satisfactory quality checks. Analysis used Python 3.12.3, NumPy 1.26.4, pandas 2.3.3, SciPy 1.11.4 and Matplotlib 3.6.3.

4 Results

4.1 Processor demand across transport configurations

SHM used less camera-process CPU than its adjacent UDP execution in every complete technical pair. Absolute savings ranged from 9.47 to 20.02 percentage points of one core. Relative savings ranged from 11.19% to 26.77% of the corresponding UDP measurement. Figure 1a shows the run means.

Batch V averaged 29.41% CPU for SHM and 39.77% for UDP, an observed difference of 10.36 percentage points. This corresponds to approximately 0.104 core-equivalent CPU time at the recorded operating point. System means were 114.69% and 124.80% on the same one-core scale. The camera probe itself averaged 18.00% in SHM and 12.50% in UDP. RouDi demand was approximately 0.08% and 0.07%, respectively. Thus, observer cost was material and was not constant across the configurations.

All ten bracketed UDP observations also exceeded their interpolated SHM references, by 8.06–15.75 points (Supplement S11). This preserves the direction under locally linear drift without resolving nonlinear or mode-dependent carryover. Cross-batch operating points remain descriptive because software, session history and background load changed together.

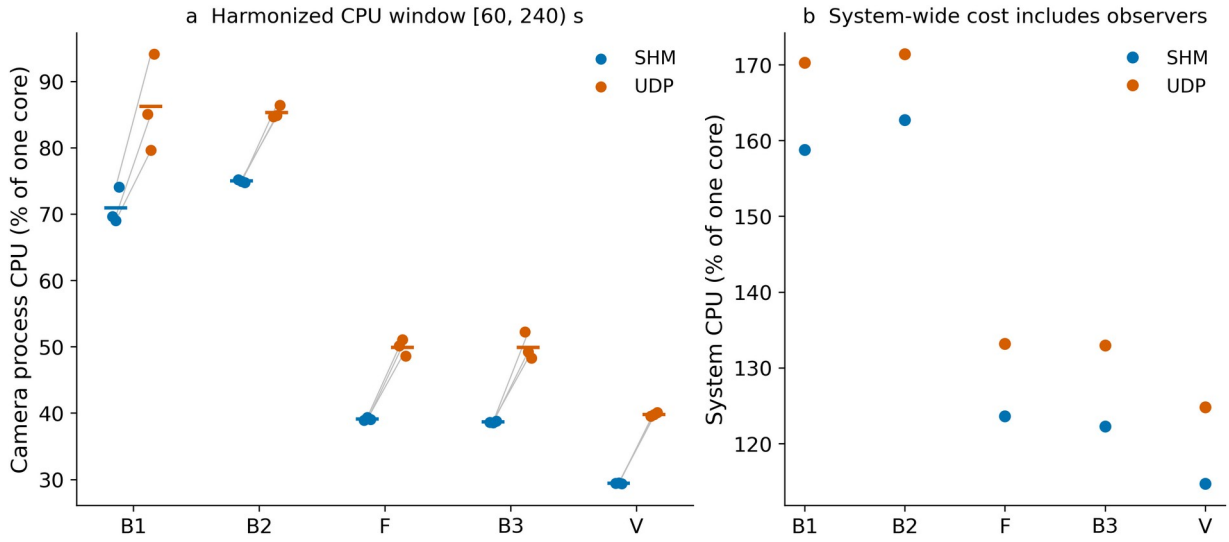


Figure 1. Camera-process and system CPU in five complete historical batches. Panel a shows three run means per transport, with horizontal cell means and links between adjacent technical pairs. Panel b shows unconnected system means including observers and background processes. CPU uses a one-core scale. Batches are observational strata; lines do not identify a causal software effect. Source: Authors' analysis of retained operational records.

4.2 Freshness and delivery behavior

CPU savings did not imply uniformly younger raw frames. In every batch, the arithmetic mean of the 15 role/run raw-frame p95 values was higher for SHM than UDP. In V, these means were 3.82 and 3.38 ms. The corresponding worst role/run p95 values were 4.66 and 3.96 ms, respectively (Figure 2

and Table II). These differences characterize the instrumented configuration, including its asymmetric probe cost.

Across complete runs, raw-image rates ranged from 29.83 to 30.00 Hz. This near-constant rate coexisted with different frame-age distributions. A separate capture-loop audit found seven roughly 15 Hz records with 97.1–100% of intervals flagged by a 50 ms rule (Supplement S10). Such flags do not independently verify hardware-frame loss. The mean role/run compressed-stream p95 age in V was 4.53 ms for SHM and 2.80 ms for UDP. However, the largest recorded compressed-stream age in V reached 4,151 ms. The aggregate record cannot locate that event within startup or steady operation, so favorable p95 values do not certify uniformly low latency.

Within the retained diagnostic windows for V, record, encoder and inference drop-counter increments were zero for both modes. Other batches contained nonzero record or inference increments. No counter reset was detected in the analyzed snapshots. The result is limited to the observed intervals; it does not establish loss-free recording over the entire deployment.

The V preview records decoded at 14.99–15.06 frames per second, with decoded-interval p95 values of 67.65–68.21 ms. Exact trace joins matched 322,550 of 322,660 decoded frames across B2, F and V (18 runs; 90 role/run records). All 1,080 recomputed archived delay statistics agreed exactly. After startup trimming, V role/run capture-marker-to-decoded p95 values ranged from 70.04 to 70.51 ms, while a 547.38 ms tail event remained at 210.49 seconds. Thus, approximately 15 Hz cadence and low ROS receiver age do not imply equally low preview age (Figure S2).

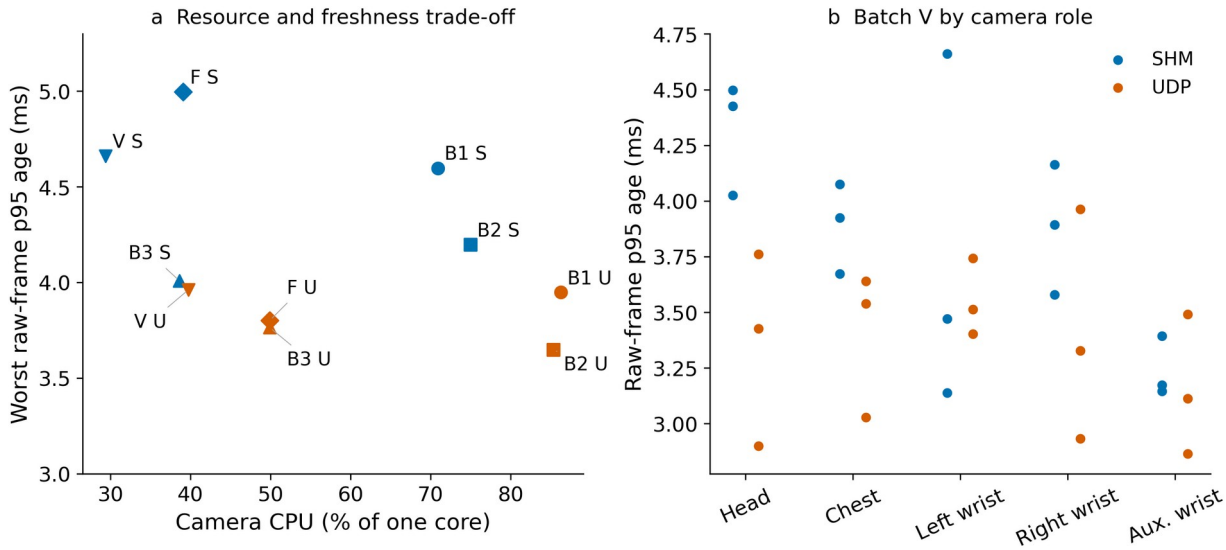


Figure 2. CPU and receiver frame freshness. Panel a pairs each cell's mean camera-process CPU with its maximum recorded raw-frame p95 across five roles and three runs. S and U denote SHM and UDP. Panel b retains the dependent role/run measurements in V. Age aggregates cover the full probe window, including startup. Source: Authors' analysis of retained operational records.

[Table II near here]

4.3 Subscriber overhead diagnostics

With Cyclone DDS verified, raw-only subscriber CPU was 14.00% for SHM and 11.55% for UDP; raw-plus-encoded values were 14.92% and 12.67%. In a separate raw-only series, jointly setting MALLOC_TRIM_THRESHOLD_ and MALLOC_MMAP_THRESHOLD_ to 536,870,912 bytes changed SHM subscriber CPU from 16.71% to 3.00%, and reported minor faults from 84,574/s to 2/s. UDP changed from 8.47% to 7.71%. These single-condition observations support an allocator-

sensitive observer-overhead hypothesis (Figure S3). They do not establish a causal allocation mechanism or optimized deployment: both thresholds changed, debugger pauses affected delivery, and publisher CPU was not measured concurrently in this series.

4.4 Window and stability audits

All 30 primary archived CPU means were reproduced to the reported 0.01-point precision. One contextual profile run differed by 0.03 points and was not used in the complete-batch comparison. Restricting the analysis to a common endpoint raised some CPU estimates because the original summaries included lower-load teardown samples. Across primary runs, time weighting changed the mean by at most 0.035 points. SHM remained lower in all 90 sensitivity contrasts across six estimators. These repetitions are correlated robustness calculations, not 90 independent experiments.

RSS trajectories explain why a declining final short-run measurement can be misleading. Several records show memory falling as subscribers exit, while the common measurement interval changes relatively little (Figure 3a). A first-to-last reduction across this boundary therefore does not establish lower memory retention during sustained operation.

The longest primary stability record contained 360 scheduled samples over 3,590 seconds. Only 182 contained node/RSS observations, with the final valid observation at 1,810 seconds. The remaining 178 observations, beginning at 1,820 seconds, contained neither a node identifier nor RSS and had empty camera statistics. The retained evidence does not distinguish node termination from an observation failure. It does establish that complete one-hour process stability was not observed.

After the recorded 300-second warmup, 152 valid RSS measurements remained. Their OLS slope was 44.35 MiB/h, whereas the Theil–Sen slope was 4.26 MiB/h. A 28.25 MiB increase at 470 seconds accounted for 93.8% of the net 30.125 MiB rise over the observed post-warmup interval. Consequently, a single linear slope does not identify a continuous memory leak. The two shorter primary records also failed their historical checks (Table III).

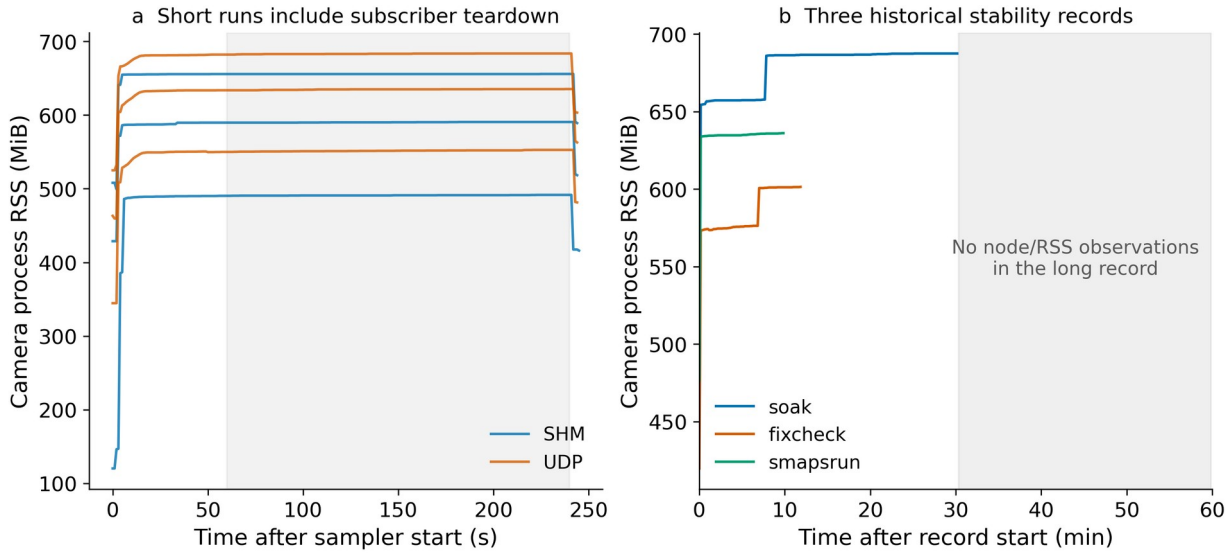


Figure 3. RSS trajectories and observation completeness. Panel a shows six V runs, with the primary CPU window shaded and memory reductions after subscribers exit. Panel b shows the three primary stability records. Shading beyond 1,820 seconds marks absent node/RSS observations in the longest record; no cause of that absence is inferred. Source: Authors' analysis of retained operational records.

[Table III near here]

Quality checks provide a separate qualification. The retained hardware-conversion outputs reported preview luma-ramp worst errors of 18 and 12 against an allowed value of 8. Recording-path error was reported as zero against an allowed value of 3. These checks use a synthetic ramp and do not measure real-scene perceptual accuracy. Nevertheless, the preview failures preclude a claim of uniformly lossless conversion.

A separate, heterogeneous archive provides two nominal 24-hour records (L1/L2), with five RGB roles, three depth streams and different SDK binaries. Both contain 2,880 probe rows spanning approximately 23.99 hours. However, L1 host monitoring ends at 15.07 hours, leaving 8.93 hours unobserved. L2 contains 2,872 host samples, a maximum boundary/sample gap of 43.02 seconds and 144/144 successful periodic eight-stream recording checks. All 85 available archived channel statistics reproduce, and four L2 monitor-source hashes match the retained run manifest. L2 satisfies the archived data, observation and periodic-recording rules, yet camera RSS increases from 630.30 to 709.07 MiB and its final status retains an RSS-growth finding (Figure S4). These records strengthen observation auditing; they neither replicate the primary transport comparison nor resolve its stability and preview-quality failures.

4.5 Conditional resource screening

An illustrative CPU budget of 40 points and a worst observed raw-frame p95 budget of 4 ms selected V/UDP as the minimum-CPU feasible cell. Relaxing the age budget to 5 ms selected V/SHM. This simple switch demonstrates why a resource-only ranking can produce a different choice from joint CPU/freshness screening. The choice also depends on aggregation. Requiring every run mean to satisfy the 40-point CPU budget makes the 4 ms case infeasible: V/UDP reached 40.06% in one run. Neither budget is a validated robot-task requirement.

Adding sustained-operation and image-quality requirements changes the interpretation again. None of the three primary stability records met its archived acceptance conditions, and preview ramp checks exceeded tolerance. L2 passes observation rules for a different workload and cannot qualify V retrospectively. The evidence therefore supports short-window screening, but does not qualify a configuration for unattended deployment.

5 Discussion

The combined results support evaluating camera-pipeline efficiency as a constrained systems problem. Transport configuration, conversion path, observer workload and measurement boundary each affect the apparent outcome. In this case, a lower CPU demand coexisted with slightly older raw frames, approximately unchanged image throughput and incomplete evidence of long-duration stability. These quantities answer different engineering questions and should remain visible together.

The transport result complements middleware and zero-copy research by exposing a deployment-level trade-off around those mechanisms. Shared ownership and reduced copying can improve resource use, but the surrounding consumers still impose costs (Bell *et al.*, 2023; Ishikawa-Aso and Kato, 2025). The subscriber diagnostics identify allocator configuration as a concrete sensitivity of probe cost. This strengthens the case for reporting observer demand separately; it does not validate adjusting primary results by a diagnostic correction factor. Privilege/profile differences remain additional explanations for part of the comparison. The data support a configuration-specific observation rather than a general ranking of SHM and UDP implementations.

Receiver frame age also adds information beyond throughput. Age-of-information theory treats timeliness as a distinct objective (Yates *et al.*, 2021), while the present metric samples age only on

arrivals. The joint screen makes this restricted metric useful without converting it into a control guarantee. The appropriate freshness limit must ultimately come from a task's dynamics, perception tolerance and decision loop. Applying a more demanding deadline could change the selected configuration even when CPU savings remain attractive.

Resource efficiency and operational qualification require different evidence. Teardown-related RSS reductions and step-dominated slopes cannot establish bounded steady-state memory. L1 further demonstrates that probe completion can coexist with incomplete host monitoring. L2 shows the complementary limit: passing observation and periodic-recording rules does not remove an RSS-growth finding. Keeping these dimensions separate prevents either favorable counters or missing observations from deciding stability alone.

Finally, computational headroom should be tested at the application level. Embedded perception studies that measure both latency and output quality provide a relevant model for that next step (Park *et al.*, 2025). A controlled follow-up would hold the policy, input recordings and power mode fixed while varying only the camera configuration. It would measure policy observation age, decision latency and task outcome together. Such a test could determine whether the measured camera savings matter to embodied execution, rather than merely shifting CPU accounting between branches.

6 Limitations

The primary comparison contains one robot, one five-role workload and technical repetitions of a shared publisher. Additional archives use different workloads and lack the primary transport controls; storage locations do not establish independent acquisition hardware. The fixed SHM-before-UDP order permits temporal and thermal confounding; the linear-drift sensitivity cannot exclude arbitrary nonlinear or mode-linked history effects. Cross-batch comparisons further combine software changes, session history and potentially background load. Missing executable hashes, historical power settings and complete device-image metadata limit replication of the original deployment, although the numerical reanalysis is reproducible.

ROS camera-age records contain aggregates: startup cannot be removed and pooled quantiles cannot be reconstructed. Matched preview traces permit a separate startup check, but do not recover ROS arrival ages. A low p95 does not bound every frame. Clock-translation uncertainty is not independently calibrated. Counter snapshots also omit intervals near window boundaries, and their branch-specific semantics differ from physical task failure.

Energy, GPU occupancy, network bandwidth, perception accuracy and learned-policy performance were not measured as primary outcomes. Synthetic conversion checks cannot replace real-scene quality evaluation. The primary stability gap prevents full-duration acceptance and does not reveal its cause; the more complete L2 archive cannot repair that missing interval. Consequently, the conclusions concern short-window resource/freshness measurements and evidence completeness, with no claim of improved autonomous task success or production readiness.

7 Conclusion

Retrospective analysis of 30 complete five-camera runs found lower camera-process CPU demand for the deployed SHM configuration in all 15 adjacent technical pairs. The saving was accompanied by slightly higher mean role/run raw-frame p95 age, showing that CPU efficiency and receiver freshness need joint constraints. Harmonized measurement windows, cumulative-counter increments and explicit missingness materially changed the interpretation of favorable summaries. Matched preview traces and subscriber diagnostics further show why observer cost and timing boundaries

must accompany resource rankings. The source tables support short-window screening. Additional long-duration records distinguish completed data probes, host observation coverage and RSS growth, while primary stability and quality failures leave sustained deployment unqualified.

Data and code availability

The supplementary reproducibility package contains anonymized source tables, analysis scripts and checksums for the reported numerical analyses. Original operational records contain internal system identifiers and are retained separately; distribution requires data-owner permission. No public repository identifier or external archival deposit is claimed.

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Table I. Cohorts in the primary archive and provenance boundaries

Cohort	Records	Retained evidence	Interpretation
B1 and B2	12	CPU series; camera/probe aggregates; counters; decoded timestamps	Separate historical baseline labels; exact binary hashes absent
F and B3	12	Same outcome families; F associated install record reports 0.45.91	B3 label does not verify equivalence to B1/B2
V	6	Same outcomes; associated 0.46.90 install and VIC/NVMM checks	Hardware-conversion configuration; cross-batch changes remain confounded
Other short records	9	Incomplete batches, profiling and diagnostic variations	Context and sensitivity; excluded from main 3+3 design
Stability records	3	RSS/process samples and historical pass/fail reasons	Different durations and observation completeness; no independent hardware replication
Subscriber diagnostics	13	8 Cyclone DDS conditions; 5 Fast DDS conditions retained separately	One execution per condition; debugger interference; not primary repetitions

All 42 primary-archive summaries are accounted for. The 30 main runs contain 15 adjacent technical pairs and 150 dependent role/run records. Hardware identification at retrieval does not establish historical power or clock settings. Source: Authors' evidence audit. Matched preview traces cover B2, F and V (18 of the 30 main runs).

Table II. Resource and freshness summaries for the complete batches

Batch	Mode	CPU mean (run range) %	Worst raw p95 ms	Mean encoded p95 ms
B1	SHM	70.90 (69.00–74.07)	4.60	7.75
B1	UDP	86.25 (79.60–94.08)	3.95	6.73
B2	SHM	74.98 (74.78–75.19)	4.20	7.56
B2	UDP	85.32 (84.66–86.43)	3.65	6.46
F	SHM	39.11 (38.93–39.33)	5.00	5.37
F	UDP	49.92 (48.61–51.04)	3.80	3.48
B3	SHM	38.65 (38.53–38.77)	4.01	4.65
B3	UDP	49.91 (48.28–52.21)	3.76	3.30
V	SHM	29.41 (29.33–29.46)	4.66	4.53
V	UDP	39.77 (39.52–40.06)	3.96	2.80

CPU is measured over [60, 240) seconds on a one-core scale, with three run means per cell. Worst raw p95 is the maximum of 15 dependent role/run p95 aggregates. Mean encoded p95 is their arithmetic mean for the compressed stream, not a pooled quantile. Age aggregates retain startup. Source: Authors' reanalysis.

Table III. Primary stability evidence at each recorded warmup cutoff

Record	Scheduled span s	Valid / planned after warmup	Observed span s	OLS / Theil–Sen MiB/h	Archived result
soak	3590	152/330	1510	44.35 / 4.26	FAIL
fixcheck	710	60/60	590	234.98 / 177.53	FAIL
smapsrun	590	54/54	530	12.10 / 11.84	FAIL

Warmup cutoffs are 300, 120 and 60 seconds, respectively. Valid counts concern RSS. The longest record has 178 missing post-warmup observations and no node identifier from 1,820 seconds onward. Slopes describe observed memory behavior rather than proving a leak mechanism. All three records failed historical acceptance checks. Source: Authors' reanalysis.